

New Products

COMPONENTS

RF/microwave capacitors

AVX Corporation has introduced a high-temperature capacitor series that provides low equivalent series resistance (ESR) and can handle high RF currents in applications like microwave RF/IF amplifiers, low-noise amplifiers, mixers, oscillators, filter networks and MRI systems.



The MK series also offers high Q values combined with a wide capacitance range of 0.1 pF to 1000 pF and voltage ratings of 50V DC to 500V DC.

Exhibiting characteristics that include low loss, electrical stability and 0 ± 30 ppm temperature coefficient, the RoHS-compliant MK series capacitors have an operating temperature range of -55°C to $+125^{\circ}\text{C}$.

AVX/Kyocera India Liaison Office, Bengaluru
Fax: (080) 41467424
Phone: 64500715
Web: www.avx.com

Low-noise amplifiers for GPS

Maxim Integrated Products has introduced the MAX2686 and MAX2688 low-noise amplifiers (LNAs) for GPS-enabled applications operating in the

1575MHz band.

Packaged in a $0.86\times 0.86\times 0.4\text{mm}^3$ -pitch wafer-level package with only four pins, the new LNAs minimise solution footprint for today's continually



shrinking handheld designs. These require only two external components to complete the board-level design (plus an optional resistor for logic-enabled shutdown) and are ideal for GPS L1, Galileo and GLONASS applications like cellphones.

The LNAs feature an ultra-low 0.75dB noise figure that improves GPS receiver sensitivity over discrete and highly integrated CMOS solutions. Additionally, customers can expect longer battery life in their end equipment, thanks to the low 4mA (typ.) operating supply current.

Maxim India, Bengaluru
Fax: (080) 30285812
Phone: 30285800
Web: www.maxim-ic.com

10Gbps repeaters

National Semiconductor has announced three multi-channel PowerWise 10Gbps repeaters that are said to provide the industry's highest equalisation gain (36 dB), consume half the power (55 mW/channel) and deliver twice the reach (20 metres over 24-AWG cable) of current industry solutions.

Powered by the company's third-generation SiGe BiCMOS process, the repeater family extends interconnect



reach and enables higher data bandwidth of up to 10.3125 Gbps. Target applications include high-speed active cable assemblies and FR-4 backplanes using serial protocols such as 10 GbE, Fibre Channel, XAUI, CPRI and Infiniband. The repeaters also support SAS/SATA out-of-band signaling for storage applications.

National Semiconductor India Designs Pvt Ltd, Bengaluru
Fax: (080) 25587921
Phone: 25587918-20, 40792200
Web: www.national.com

High-speed 8-bit ADCs

Analog Devices, Inc. (ADI) has released two 8-bit, high-speed, low-power, low-noise analogue-to-digital converters (ADCs) for instrumentation and communication applications. The new ADCs use a pipeline architecture and consume just 310 mW of power.

The AD9284 is claimed to be the industry's first dual, 8-bit, 250MSPS ADC, while the AD9286 8-bit 500MSPS ADC delivers 8-bit resolution at 40 per cent less power than competing components. The converters feature high signal-to-noise ratio of 49.3 dBFS and allow for a wide dynamic signal range with spurious-free dynamic range performance of 65 dBc. Applications include battery-powered instruments, handheld scope meters and digital oscilloscopes.

Analog Devices India Pvt Ltd, Bengaluru
Fax: (080) 43002333
Phone: 43002000
Web: www.analog.com